

Shyam Hemamalini

Computational Mechanical Engineer

CFD · Flow&Thermal · Mechanical design · Code development

+31-6-49490069 | shyam.hemamalini@gmail.com | [LinkedIn](#)

Nationality: Dutch | Availability: August 2026



PROFESSIONAL SUMMARY

Highly skilled Computational Engineer and high-impact R&D engineer with a strong foundation in first-principles physics and extensive experience developing end-to-end mechanical, flow&thermal, energy systems. A dedicated researcher with a distinct industrial outlook, adept at leveraging high-performance computing (HPC) alongside a robust suite of commercial and open-source software packages to drive product innovation from concept to validation across mechanical engineering. Proven experience in bridging the gap between advanced multi-physics software development and practical industrial R&D.

RESEARCH & CAREER INTERESTS

- **Multiphysics Modelling & Simulation** – Advanced expertise including but not limited to structural mechanics, fluid dynamics (CFD), heat transfer, and coupled Euler-Lagrange systems. Proficient in Ansys Workbench, Ansys Fluent, and OpenFOAM.
- **High-Performance Computing (HPC) & Software Development** – Experienced in architecting, testing, and implementing high-fidelity simulation software using C/C++, Python, Fortran, MATLAB, and CUDA for GPU-accelerated computing.
- **Mechanical Design & Control Systems** – Proficient in SolidWorks/Blender for CAD generation, MATLAB Simulink for structural analysis, Gmsh/snappy/Ansys Meshing with a deep analytical toolkit for mesh optimization, spectral and time analysis (PSD), and MATLAB/Simulink/C# for modeling complex kinematics, robot arm movements, and closed-loop actuator control loops.

TECHNICAL SKILLS

Languages: Python, Fortran, MATLAB, C/C++, C#, CUDA, Bash

Tools: Ansys Fluent, OpenFOAM, Solidworks, Blender, Paraview, Adobe Suite, Git, VS Code, Simulink

Research Areas: Multiphase flows, turbulent flows, species transport, chemically reacting flows, heat & mass transfer, mechanical design, solid dynamics

EDUCATION & EXPERIENCE

PhD Candidate in Mechanical Engineering

Technische Universiteit Eindhoven

July 2022 – July 2026 (expected)

Eindhoven, Netherlands

Dissertation: Why do iron particles fail to burn? Numerical investigations into the interaction of turbulence and iron powder combustion.

Advisors: Dr. XiaoCheng Mi, prof.dr.ir. Jeroen van Oijen, Dr. Benedicte Cuenot

- **Developed a hybrid multiphysics solver** for turbulent iron powder flame simulations with a high-fidelity Fortran DNS framework (NTMIX-CHEMKIN) and a custom-built C++ OpenFOAM LES solver using a four-way Euler-Lagrange approach.
- **Engineered a scalable Python postprocessing framework** with scipy, h5py, pandas for CFD post-processing, optimizing memory management and parallel execution to efficiently handle 500GB+ datasets.
- **Designed and executed research-level simulations** that led to valuable analytical insights and actionable conclusions on flow-particle interactions in large-scale burners.
- Submitted 5 papers to top journals (PROCI, CnF, Fuel), presented oral presentations and posters at over 10 conferences, and **won the Best Poster Award at COMBURA 2024.**

Research Assistant

Wageningen University & Research

Feb. 2022 – Dec. 2022

Wageningen, Netherlands

- **Modeled transient indoor airborne virus transmission dynamics** using Ansys Fluent to evaluate and quantify COVID-19 infection risks across various architectural scenarios.
- **Developed and implemented a custom User-Defined Function (UDF) plugin** in C to accurately simulate the multiphase, high-velocity fluid dynamics of human coughing and sneezing events.
- **Translated complex CFD data into actionable epidemiological insights**, providing a quantitative framework to optimize indoor safety protocols and ventilation design.

Master of Science in Mechanical Engineering

Technische Universiteit Delft

Aug. 2019 – December 2021

Delft, Netherlands

Specialization: Energy, Flow, and Process Engineering

Thesis Title: [A Lagrangian passive scalar solver for mass transport in electrolytes and coupling to the particle-resolved solver Bluebottle](#) (supervised by Dr. Lorenzo Botto)

- **Developed a novel C/CUDA-based Lagrangian solver** based on the Nernst-Planck equation to model multi-physics ionic flux, integrating it into the open-source multiphase solver Bluebottle.
- **Utilized high-performance parallelized computing** to simulate electrokinetic transport in water electrolyzers, successfully isolating the impact of gas bubble impedance on overall current density.
- **Performed rigorous validation against physical experimental data**, proving the framework's efficacy as an industrial-grade predictive modeling tool.

Intern – MSc Internship

TNO - Heat Transfer & Fluid Dynamics group

Aug. 2020 – Oct. 2020

Delft, Netherlands

- **Modeled complex multiphase liquid-gas flows** through flexible and curved piping geometries in OpenFOAM and Ansys Fluent; generated high-fidelity structured and unstructured meshes optimized for boundary layer resolution and phase-interface tracking.
- **Developed automated MATLAB post-processing pipelines** to analyze transient hydrodynamic force magnitudes and compute Power Spectral Density (PSD) data, successfully identifying flow-induced vibration frequencies.
- **Validated computational fluid dynamics (CFD) models against in-house experimental data**, performing rigorous error-quantification and spectral analysis to ensure simulation accuracy across varying pipe configurations.

Junior Research Fellow

PSG College of Technology

Jul. 2018 – Jun. 2019

Coimbatore, India

- **Developed analytical multiphysics models from first principles** to simulate closed-loop thermosiphon cooling, balancing buoyancy forces against viscous resistance to establish natural fluid circulation without mechanical pumps.
- **Built an iterative transient simulation model** with 30-second time-step discretization to map daily solar irradiance profiles, track module heat capacities, and quantify daily cumulative energy losses.
- **Engineered and optimized a parallel-plate fin heat sink manifold** using SolidWorks, applying Teertstra's analytical equations to achieve 98.5% fin efficiency and validating computational models against physical experimental data.
- Project report link: [Numerical simulation of natural convection along asymmetrically heated inclined open ended channels with steep inclination and high aspect ratios](#)

Bachelor of Engineering in Mechanical Engineering

Coimbatore Institute of Technology

Aug. 2014 – May 2018

Coimbatore, India

Specialization: Mechanical design, thermodynamics, fluid mechanics and control systems

End Project Title: Fabrication of foldable portable cycle for optimized frame design

- **Geometrically optimized and fabricated a novel Y-frame folding bicycle**, formulating a multivariate non-linear objective function and using pattern search methods to establish an optimal 72° head angle.
- **Developed 3D CAD models and executed structural finite element analysis (FEA)**, validating full-vehicle integrity under extreme 120 kg loads with strict factors of safety (FOS 2–3).
- **Engineered custom hardware mechanisms, including telescoping forks and notch-key sliding locks**, to achieve compact portability, subsequently validating the physical prototype through real-time endurance testing.
- Visualization: [3D CAD model from SolidWorks](#)

Design Engineer

Team Drifting Dragons

May 2016 – Feb. 2017
Coimbatore, India

- **Led the design department for a multidisciplinary BAJA SAE India team**, overseeing the end-to-end CAD engineering, full-vehicle system integration (powertrain, suspension, steering), and assembly of an all-terrain racing vehicle.
- **Engineered and optimized the structural roll cage using SolidWorks for 3D modeling**, ensuring optimal ergonomics, driver safety, and seamless tolerance stack-up troubleshooting during physical assembly.
- **Executed advanced structural solid dynamics simulations (FEA) in Ansys Workbench**, conducting static, impact, and torsional stiffness analyses to optimize wall thickness, validate welds, and pass regulatory safety tests with excellent ratings.
- Visualization: [3D CAD model from SolidWorks](#)

Participant in eYantra Robotics Competition

Coimbatore Institute of Technology

Jun. 2015 – Apr. 2016
Coimbatore, India

- **Designed, assembled, and programmed an autonomous line-following and waste-sorting robot** for an IIT Bombay national competition, implementing navigation algorithms driven by color and ambient light sensor feedback loops.
- **Developed embedded firmware in C# for ATmega microcontrollers**, establishing low-level control loops for real-time sensor processing and precise high-torque servo motor actuation.
- **Utilized MATLAB/Simulink to model and simulate robot arm movements and grip control**, optimizing transient response and kinematic stability prior to physical hardware deployment.

PUBLICATIONS

- | | |
|---|------|
| 1. <i>The influence of preferential concentration on the ignition and combustion of iron particles in a spherical suspension</i>
In preparation
Shyam Hemamalini , Lotte Meeuwsen, XiaoCheng Mi | 2026 |
| 2. <i>LES of iron-powder combustion in a jet-in-hot-coflow burner – Insights on flame structure and ignition characteristics</i>
Proc. Combustion Symposium (accepted)
Shyam Hemamalini , XiaoCheng Mi | 2026 |
| 3. <i>Effects of preferential concentration on the combustion of iron particles – A numerical study with homogeneous isotropic turbulence</i>
Combustion & Flame DOI:10.1016/j.combustflame.2026.115119
Shyam Hemamalini , Benedicte Cuenot, XiaoCheng Mi | 2026 |
| 4. <i>A theoretical analysis of timescales in preferential concentration of burning iron particles</i>
Fuel - Short Communication (in review) Preprint: DOI:10.2139/ssrn.5138232
Shyam Hemamalini , Benedicte Cuenot, XiaoCheng Mi | 2026 |
| 5. <i>Numerical study probing the effects of preferential concentration on the combustion of iron particles in a mixing layer</i>
Proc. Combustion Symposium DOI:10.1016/j.proci.2024.105617
Shyam Hemamalini , Benedicte Cuenot, Jeroen van Oijen, XiaoCheng Mi | 2024 |

ANCILLARY ACTIVITIES

- | | |
|---|-------------|
| Teaching Assistant
4RC30 - Introduction to Computational Fluid Dynamics at TU Eindhoven | 2023 – 2025 |
| Teaching Assistant
MOOC - <i>“Technology for Intelligent and Integrated Energy Systems”</i> on edX | 2021 – 2023 |
| Volunteer Teacher
High-school level physics and chemistry at Abdul Kalam Institute for IIT-JEE, Coimbatore, India | 2018 – 2019 |